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Recursion, bijections and double counting

Level 4 Olympiad · Combinatorics

Three ways to count cleverly: count one set two ways, match it with something easier, or build it from smaller cases, with the rigour a BMO proof needs.

By the end you can: You will be able to set up and justify recursions, using states when one sequence is not enough.; You will be able to construct bijections and prove them by writing down the inverse.; You will be able to double count incidences and pairs to find exact answers and to prove bounds..

Lesson

Twelve students each belong to exactly three clubs, and every club has exactly five members. How many clubs are there?

Count the pairs (student, club) with the student in the club. Counting by students gives $12 \times 3 = 36$. Counting by clubs gives $5c$ if there are c clubs. So $5c = 36$, which is impossible: no such arrangement exists. Two lines, no casework, and a fact that would be painful to discover by trying to build the clubs.

This unit is about three ways of counting cleverly: count one set in two ways (double counting), match it with a set that is easier to count (bijections), and build it from smaller cases (recursion). Most AIME counting questions are one of these three ideas in disguise, and BMO combinatorics problems often turn on one of them.

Double counting

Double counting. Let A and B be finite sets and R a set of pairs (a, b) with $a \in A$ and $b \in B$. If $r(a)$ is the number of pairs in R containing a, and $c(b)$ the number containing b, then

$$\sum_{a \in A} r(a) = |R| = \sum_{b \in B} c(b).$$

Both sides count R, sorted by first entry or by second entry. Two consequences you should know:

- **Handshake lemma.** In any group, the sum over people of the number of friends each has equals twice the number of friendships, since each friendship is counted from both ends. In particular the number of people with an odd number of friends is even.
- **Committee identities.** Count committees of k people chosen from n, with a chair: choose the committee and then its chair, or the chair and then the other $k-1$ members. So $k\binom{n}{k} = n\binom{n-1}{k-1}$.

The art is choosing what to count. Usually it is incidences (a person on a committee) or pairs (two people on the same committee).

Example 1. Ten people form some committees of four. Every two of the ten people serve together on exactly two committees. How many committees are there, and on how many committees does each person serve?

Solution.
Counting incidences does not work straight away, since we know neither the number of

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committees nor how many each person is on. The information we are given is about pairs, so count pairs.
Count the pairs (two people, committee) where both people are on the committee. By pairs of people: there are $\binom{10}{2} = 45$ of them, each on two committees, giving 90. By committees: each committee contains $\binom{4}{2} = 6$ pairs of people, giving $6c$ if there are c committees. So $c = 15$.
Now fix a person P and count the pairs (another person Q, committee containing both P and Q). By Q: $9 \times 2 = 18$. By committee: each of the r committees containing P contains 3 other people, giving $3r$. So $r = 6$, the same for everyone. As a check, the incidences come to $15 \times 4 = 60 = 10 \times 6$.

Such a system of committees does exist, but the counting gave the numbers without our having to build one.

Bijections

Bijection principle. If $f : A \to B$ has an inverse, that is, a map $g : B \to A$ with $g(f(a)) = a$ and $f(g(b)) = b$ for all $a \in A$ and $b \in B$, then $|A| = |B|$.

In practice you describe the map, describe how to undo it, and check both directions; that is the proof. The bijections you should know by heart:

- subsets of $\{1, \ldots, n\}$ and binary strings of length n, so there are 2^n of each;
- paths of unit steps right and up from $(0, 0)$ to (m, n), and words with m letters R and n letters U, so there are $\binom{m+n}{m}$ of each;
- stars and bars, from L3-CO1: solutions of $x_1 + \cdots + x_k = n$ in non-negative integers and rows of n stars and $k - 1$ bars.

Example 2. A row in a cinema has 20 seats. In how many ways can 5 people choose seats (we only care which seats are taken) so that between any two of them there are at least two empty seats?

Solution.
Casework on the sizes of the gaps is a mess. Write the chosen seat numbers as $s_1 < s_2 < \cdots < s_5$; the condition is $s_{i+1} - s_i \geq 3$. The trick is to squeeze out the empty seats that are forced. Set

$$t_i = s_i - 2(i - 1),$$

so $t_1 = s_1$, $t_2 = s_2 - 2$, and so on up to $t_5 = s_5 - 8$.
Then $t_{i+1} - t_i = s_{i+1} - s_i - 2 \geq 1$, so the t_i increase, with $t_1 \geq 1$ and $t_5 \leq 20 - 8 = 12$. So $\{t_1, \ldots, t_5\}$ is a 5-element subset of $\{1, \ldots, 12\}$. Conversely, any 5-element subset $t_1 < \cdots < t_5$ of $\{1, \ldots, 12\}$ gives $s_i = t_i + 2(i - 1)$, which satisfies $s_{i+1} - s_i = t_{i+1} - t_i + 2 \geq 3$ and $s_5 \leq 20$. The two maps undo each other, so the answer is $\binom{12}{5} = 792$.

The general version: the number of k-element subsets of $\{1, \ldots, n\}$ in which any two elements differ by at least d is $\binom{n-(k-1)(d-1)}{k}$.

Recursion

When the objects have a natural size n, let a_n be the number of objects of size n and ask how an object of size n breaks up into a smaller object plus a last piece. Splitting by the last piece (or the first) gives an equation for a_n in terms of earlier terms. If the pieces you may add depend on what

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came before, one sequence is not enough: keep track of the possible endings as separate sequences, often called states.

Making a recursion rigorous. A recurrence such as $a_n = a_{n-1} + a_{n-2}$ is a bijection in disguise. It says that the objects of size n split into two disjoint classes, one matched with the objects of size $n-1$ and the other with the objects of size $n-2$. In a proof, say what the classes are, why they do not overlap, why they cover everything, and what the matching is. Then check the first few values by hand.

Example 3. How many strings of length 8 made from the letters A, B and C do not contain AB as two consecutive letters?

Solution.
The complement looks attractive: count the strings with at least one AB and subtract from 3^8. But AB can occur in several places, and occurrences can sit side by side (as in ABAB), so inclusion-exclusion becomes a mess. Recursion is cleaner.
Whether we may add a B depends only on whether the string so far ends in A. So let x_n be the number of good strings of length n ending in A, let y_n be the number ending in B or C, and let $t_n = x_n + y_n$. A good string of length $n+1$ ending in A is any good string of length n followed by A, so $x_{n+1} = t_n$. A good string of length $n+1$ ending in B or C is either a good string ending in A followed by C, or a good string not ending in A followed by B or C, so $y_{n+1} = x_n + 2y_n$.
Then $t_{n+1} = t_n + x_n + 2y_n = 3t_n - x_n = 3t_n - t_{n-1}$, using $x_n = t_{n-1}$. With $t_1 = 3$ and $t_2 = 9 - 1 = 8$ we get

$$3,\ 8,\ 21,\ 55,\ 144,\ 377,\ 987,\ 2584,$$

so the answer is 2584.
These are alternate Fibonacci numbers: with $F_1 = F_2 = 1$, we have $t_n = F_{2n+2}$. That is no accident, since $F_{m+2} = 3F_m - F_{m-2}$ for every $m \geq 3$, but you did not need to spot it to get the answer.

Combining the ideas

Harder problems mix the three. The next example uses double counting for a bound and a bijection for the construction, and it shows that the choice of what to count can matter a great deal.

Example 4. Some cells of a 5×10 grid (5 rows and 10 columns) are coloured so that no four coloured cells are the corners of a rectangle with sides along grid lines. What is the largest possible number of coloured cells?

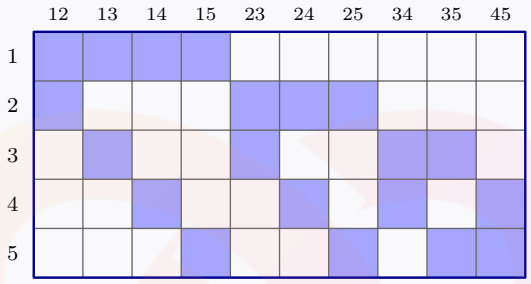
Solution.
A rectangle is two rows and two columns with all four crossing cells coloured. So "no rectangle" says that two columns never share two coloured rows, or equivalently that two rows never share two coloured columns. Let N be the number of coloured cells.
First attempt: count pairs of coloured cells in the same row. If row i has r_i coloured cells, it uses $\binom{r_i}{2}$ pairs of columns, and no pair of columns may be used by two rows, so $\sum \binom{r_i}{2} \leq \binom{10}{2} = 45$. With 24 cells in 5 rows the left side is at least $4\binom{5}{2} + \binom{4}{2} = 46$ (the most even split is $5, 5, 5, 5, 4$), so this gives $N \leq 23$. That needs a smoothing argument to be rigorous, and 23 turns out to be far from the truth.

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Count the other way. If column j has c_j coloured cells, it uses $\binom{c_j}{2}$ pairs of rows, and there are only $\binom{5}{2} = 10$ pairs of rows to go round. So

$$\sum_{j=1}^{10} \binom{c_j}{2} \leq 10.$$

Now $\binom{c}{2} - (c-1) = \frac{(c-1)(c-2)}{2} \geq 0$ for every integer $c \geq 0$, so $\binom{c_j}{2} \geq c_j - 1$. Adding over the ten columns gives $N - 10 \leq 10$, so $N \leq 20$.
For equality every column must contain exactly two coloured cells and every pair of rows must be used exactly once. That is a bijection between the 10 columns and the 10 pairs of rows: in each column, colour the two rows of its pair. Two columns then share at most one coloured row, so there is no rectangle, and we have 20 coloured cells. The answer is 20.



Counting pairs from the side with fewer possible pairs (here, pairs of rows) gave the sharper bound. When one double count is weak, try the other.

Common traps.

- Writing down a map and calling it a bijection without checking that it can be reversed. Give the inverse map explicitly, and check that each map lands in the right set.
- Off-by-one errors in recursions. Check a_1 and a_2 (and a_0, if you use it) by hand before trusting the recurrence.
- Classes in a recursion that overlap or miss cases. Split by something definite, such as the last letter or the length of the last block.
- Ordered versus unordered pairs in double counting: a friendship is one unordered pair but two ordered ones.
- Fibonacci indexing. Say which convention you are using.

How to spot this in a contest

Phrases such as "each ... exactly k", "every two ... at most one" or "the sum over all subsets" point to double counting: count incidences or pairs, from both sides. If you need a bound, look for an inequality such as $\binom{c}{2} \geq c - 1$, or an "as equal as possible" argument, to finish.

If small cases give binomial coefficients, powers of 2 or Fibonacci numbers, there is probably a bijection or a recursion behind them, so compute small cases first. Forbidden patterns ("no two adjacent", "no AB", "no three in a row") almost always mean a recursion with states. On the AIME the recurrence gets you a number; on the BMO you must also justify it, by stating the classes and the matching.

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Practice

Try each problem before turning to the hints and solutions at the back.

1. ⋆ In how many ways can a 1×12 strip be tiled with 1×1 squares and 1×3 rectangles?

2. ⋆⋆ For each non-empty subset of $\{1, 2, \ldots, 10\}$, write down its largest element. Find the sum of all the numbers written down.

3. ⋆⋆ How many strings of ten binary digits contain the block 01 (a 0 immediately followed by a 1) exactly three times?

4. ⋆⋆⋆ In a club, every member has exactly 5 friends among the other members (friendship is mutual). No two friends have a common friend, and any two members who are not friends have exactly 2 common friends. How many members does the club have?

5. ⋆⋆⋆ Let n be a positive integer. Call a subset of $\{1, 2, \ldots, 2n\}$ *balanced* if it contains as many odd numbers as even numbers (the empty set is balanced). Prove that the number of balanced subsets is $\binom{2n}{n}$.

6. ⋆⋆⋆⋆ In a group of nine people, every two are either friends or strangers. Call a set of three people *uniform* if the three are mutual friends or mutual strangers. Prove that there are at least 12 uniform sets.

Contest ladder

Real past problems that use this idea, easiest first. Open them on the official paper or the AoPS wiki, or online at kevinvariant08.github.io.

1. AIME I 2023 Problem 3 ⋆
2. BMO Round 1 2013/14 Problem 4 ⋆⋆
3. AMC 10B 2019 Problem 25 ⋆⋆
4. BMO Round 1 2023/24 Problem 1 ⋆⋆
5. BMO Round 1 2010/11 Problem 4 ⋆⋆
6. AIME I 2017 Problem 7 ⋆⋆⋆
7. BMO Round 1 2017/18 Problem 2 ⋆⋆⋆
8. AIME I 2018 Problem 10 ⋆⋆⋆
9. AIME I 2015 Problem 12 ⋆⋆⋆
10. AIME I 2022 Problem 12 ⋆⋆⋆⋆
11. BMO Round 1 2018/19 Problem 6 ⋆⋆⋆⋆

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Hints

1. Look at the tile covering the rightmost cell.
2. Instead of going subset by subset, ask for each k how many subsets have largest element k, or how many have largest element at least j.
3. Put an extra 1 at the front and an extra 0 at the end. Now count the places where the digit changes.
4. Fix one member X and count the paths X, Y, Z in which Y is a friend of X and $Z \neq X$ is a friend of Y.
5. Keep the odd numbers of a balanced subset as they are, but swap its even part for its complement among the even numbers.
6. For a set of three that is not uniform, look at each of its members and ask whether that person is friends with exactly one of the other two. Count such "mixed corners" in two ways.

Solutions

Problem 1

Answer: 60

Let a_n be the number of tilings of a $1 \times n$ strip. The rightmost cell is covered either by a square, leaving a $1 \times (n-1)$ strip, or by a 1×3 rectangle, leaving a $1 \times (n-3)$ strip. These cases do not overlap, so $a_n = a_{n-1} + a_{n-3}$ for $n \geq 3$, with $a_0 = a_1 = a_2 = 1$. Then

$$a_3, a_4, \ldots, a_{12} = 2,\ 3,\ 4,\ 6,\ 9,\ 13,\ 19,\ 28,\ 41,\ 60,$$

so there are 60 tilings.

Problem 2

Answer: 9217

Write the largest element m of a subset as $m = \sum_{j=1}^{10} [m \geq j]$, where $[m \geq j]$ is 1 if $m \geq j$ and 0 otherwise. Swapping the order of summation, the required total is

$$\sum_{j=1}^{10} \#\{\text{non-empty subsets with largest element at least } j\}.$$

A subset has largest element less than j exactly when it is a subset of $\{1, \ldots, j-1\}$, so the number with largest element at least j is $2^{10} - 2^{j-1}$. The total is

$$\sum_{j=1}^{10} \left(2^{10} - 2^{j-1}\right) = 10 \cdot 1024 - (2^{10} - 1) = 10240 - 1023 = 9217.$$

(Alternatively, k is the largest element of 2^{k-1} subsets, and $\sum_{k=1}^{10} k\, 2^{k-1} = 9 \cdot 2^{10} + 1 = 9217$.)

Problem 3

Answer: 330

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Given such a string s, form the 12-digit string s' by putting a 1 in front of s and a 0 after it. The new adjacent pairs at the ends read $1x$ and $x0$, so neither is 01, and s' contains 01 exactly as often as s does.

In s', look at the 11 gaps between consecutive digits and mark those where the digit changes. Since s' starts with 1 and ends with 0, the changes alternate $1 \to 0$, $0 \to 1$, $1 \to 0$, and so on, starting and finishing with a $1 \to 0$ change. So if there are three $0 \to 1$ changes there are four $1 \to 0$ changes: seven changes in all. Conversely, any choice of 7 of the 11 gaps, together with the first digit 1, determines a 12-digit string starting with 1, ending with 0 (an odd number of changes) and containing exactly three blocks 01; deleting its first and last digits gives a valid s. These maps undo each other, so the answer is $\binom{11}{7} = 330$.

Problem 4

Answer: 16

Fix a member X and count the pairs (Y, Z) where Y is a friend of X, Z is a friend of Y, and $Z \neq X$. There are 5 choices of Y and then 4 choices of Z, so 20 pairs.

Count them by Z instead. Z cannot be a friend of X: otherwise the friends X and Y would have the common friend Z. So Z is one of the members other than X who are not friends of X, and each such member, having exactly 2 common friends with X, appears in exactly 2 pairs. Hence there are $20/2 = 10$ members who are neither X nor friends of X, and the club has $1 + 5 + 10 = 16$ members.

Such a club exists. Take as members the 16 strings of four binary digits, and make two strings friends when they differ in exactly one digit or in all four. Each string has 5 friends; one checks that friends have no common friend and that non-friends have exactly 2.

Problem 5

Let O and E be the sets of odd and even numbers in $\{1, \ldots, 2n\}$; each has n elements. For any subset X of $\{1, \ldots, 2n\}$ define

$$\varphi(X) = (X \cap O) \cup (E \setminus X),$$

that is, keep the odd elements of X and replace its even elements by the even numbers not in X.

Applying φ twice gives back X: the odd part is unchanged, and the even part is complemented twice. So φ is a bijection from the set of all subsets to itself, and it is its own inverse.

If X is balanced with k odd and k even elements, then $\varphi(X)$ has k odd elements and $n - k$ even elements, so it has exactly n elements. Conversely, if $|Y| = n$ with j odd elements, then Y has $n - j$ even elements, and $\varphi(Y)$ has j odd elements and $n - (n - j) = j$ even elements, so $\varphi(Y)$ is balanced. Therefore φ maps the balanced subsets into the n-element subsets and the n-element subsets into the balanced subsets, and since φ is its own inverse these two restrictions are inverse bijections. Hence the number of balanced subsets equals the number of n-element subsets of $\{1, \ldots, 2n\}$, which is $\binom{2n}{n}$.

(Counting balanced subsets directly by the value of k gives $\sum_{k=0}^{n} \binom{n}{k}^2$, so this also proves $\sum_k \binom{n}{k}^2 = \binom{2n}{n}$.)

Problem 6

There are $\binom{9}{3} = 84$ sets of three people. Call a pair $(P, \{Q, R\})$, where P is a person and $\{Q, R\}$ is a pair of two other people, a *mixed corner* if P is friends with exactly one of Q and R. We count mixed corners in two ways.

By sets of three. In a uniform set $\{P, Q, R\}$ no corner is mixed. In a set that is not uniform, two of the three pairs are of one type and the third pair is of the other type. The person shared by the

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two pairs of the same type is not a mixed corner, and each of the other two people is joined to the others by one pair of each type, so is a mixed corner. So each non-uniform set contributes exactly 2 mixed corners, and if there are u uniform sets the number of mixed corners is $2(84-u)$.

By people. If person P has d friends, then P has $8-d$ strangers, and P is a mixed corner for exactly $d(8-d)$ pairs. Since $d(8-d) = 16-(d-4)^2 \leq 16$, the number of mixed corners is at most $9 \times 16 = 144$.

Hence $2(84-u) \leq 144$, so $u \geq 84-72 = 12$.

(Equality is possible: arrange the nine people in a 3×3 array and make two people friends when they are in the same row or the same column. Everyone then has 4 friends, and the uniform sets are the 3 rows, the 3 columns and the 6 sets of three people in different rows and different columns.)

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